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Batch : 16

Date : 10 September 2026

1. Aim

To design, develop and implement a menu-driven C program using the switch-case construct that performs the following digit-based operations on a positive integer entered by the user, where each operation is implemented in its own separate void function:

- Generate the largest possible number using its digits.
- Generate the smallest possible number using its digits.
- Swap the first and last digits of the number.
- Check whether the number is a palindrome.

2. Algorithm

2.1 Main Menu

1. Start the program.
2. Display the menu with five options: Largest number, Smallest number, Swap first & last digit, Check palindrome, and Exit.
3. Read the user's choice.
4. Using a switch-case statement, call the corresponding function: Case 1 → generateLargest() Case 2 → generateSmallest() Case 3 → swapFirstLast() Case 4 → checkPalindrome() Case 5 → Exit the loop Default → Display "Invalid choice"
5. Repeat steps 2–4 inside a do-while loop until the user selects Exit (choice = 5).
6. Stop the program.

2.2 generateLargest() — Largest number using digits

1. Read a positive integer N.
2. Extract each digit of N (using % and /) and store the digits in an array.
3. Sort the digit array in descending order (bubble sort).
4. Print the digits in the sorted order — this forms the largest possible number.

2.3 generateSmallest() — Smallest number using digits

1. Read a positive integer N.
2. Extract each digit of N and store the digits in an array.
3. Sort the digit array in ascending order (bubble sort).
4. If the number has more than one digit and the first digit of the sorted array is 0, swap it with the first non-zero digit found (to avoid a meaningless leading zero).
5. Print the digits — this forms the smallest possible number.

2.4 swapFirstLast() — Swap first and last digit

1. Read a positive integer N and store a copy of it as the original number.
2. Extract the digits of N into an array (the first digit extracted is the last digit of N).
3. Swap the value at index 0 (last digit) with the value at the last index (first digit) of the digit array.

- Print the digits from the last index to index 0 — this gives the number with the first and last digits swapped.

2.5 checkPalindrome() — Palindrome check

- Read a positive integer N and store it as the original number.
- Initialise reverse = 0.
- Repeat while N > 0: extract the last digit (remainder = N % 10), update reverse = reverse * 10 + remainder, and update N = N / 10.
- Compare the reverse with the original number.
- If they are equal, print that the number is a palindrome; otherwise print that it is not.

3. Test Case Tables

3.1 Menu Option 1 — Generate Largest Number

Test No.	Input (N)	Expected Output	Actual Output	Result
1	3521	5321	5321	Pass
2	120	210	210	Pass
3	9	9	9	Pass
4	111	111	111	Pass

3.2 Menu Option 2 — Generate Smallest Number

Test No.	Input (N)	Expected Output	Actual Output	Result
1	3521	1235	1235	Pass
2	1002	1002	1002	Pass
3	9	9	9	Pass
4	4321	1234	1234	Pass

3.3 Menu Option 3 — Swap First and Last Digit

Test No.	Input (N)	Expected Output	Actual Output	Result
1	1234	4231	4231	Pass
2	7	7	7	Pass
3	10	01 → 1	1	Pass
4	56789	96785	96785	Pass

3.4 Menu Option 4 — Palindrome Check

Test No.	Input (N)	Expected Output	Actual Output	Result
1	121	Palindrome	Palindrome	Pass
2	1234	Not a Palindrome	Not a Palindrome	Pass
3	7	Palindrome	Palindrome	Pass
4	12321	Palindrome	Palindrome	Pass

4. Source Code

File name: digit_ops.c

```

/*****
 * Program : Digit-Based Operations on a Positive Integer
 * Author  : Rudra Pratap Singh (SAP ID: 590032811)
 * Batch   : 16
 * Date    : 10 September 2026
 *
 * Description:
 * A menu-driven C program that performs the following digit
 * based operations on a positive integer entered by the user:
 *     1. Generate the largest possible number using its digits
 *     2. Generate the smallest possible number using its digits
 *     3. Swap its first and last digits
 *     4. Check whether it is a palindrome number
 *     5. Exit
 *
 * Each operation is implemented as a separate void function.
 * Every function reads its own input, processes it and prints
 * the result (no arguments are passed to / returned from the
 * functions - i.e. no call by value / call by reference).
 *****/

#include <stdio.h>

/* ---- Function prototypes (simple calls, no parameters) ---- */
void generateLargest(void);
void generateSmallest(void);
void swapFirstLast(void);
void checkPalindrome(void);

int main(void)
{
    int choice;

    do
    {
        printf("\n===== \n");
        printf("  DIGIT BASED OPERATIONS ON A POSITIVE INTEGER\n");
        printf("===== \n");
        printf("1. Generate the largest possible number\n");
        printf("2. Generate the smallest possible number\n");
        printf("3. Swap first and last digit\n");
        printf("4. Check whether the number is a palindrome\n");
        printf("5. Exit\n");
        printf("----- \n");
    }

```

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printf("Enter your choice (1-5): ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        generateLargest();
        break;
    case 2:
        generateSmallest();
        break;
    case 3:
        swapFirstLast();
        break;
    case 4:
        checkPalindrome();
        break;
    case 5:
        printf("\nExiting the program. Thank you!\n");
        break;
    default:
        printf("\nInvalid choice! Please enter a number between 1 and 5.\n");
}

} while (choice != 5);

return 0;
}

/*****
* Function : generateLargest
* Purpose  : Reads a positive integer, arranges its digits in
*             descending order to form the largest possible
*             number and displays it.
*****/
void generateLargest(void)
{
    long long num;
    int digits[20], count = 0, i, j, temp;

    printf("\nEnter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return;
    }

    /* Extract digits */
    while (num > 0)
    {
        digits[count++] = num % 10;
        num = num / 10;
    }

    /* Sort digits in descending order (bubble sort) */
    for (i = 0; i < count - 1; i++)
    {
        for (j = 0; j < count - 1 - i; j++)
        {

```

```

        if (digits[j] < digits[j + 1])
        {
            temp = digits[j];
            digits[j] = digits[j + 1];
            digits[j + 1] = temp;
        }
    }

    printf("Largest number possible : ");
    for (i = 0; i < count; i++)
        printf("%d", digits[i]);
    printf("\n");
}

/*****
 * Function : generateSmallest
 * Purpose  : Reads a positive integer, arranges its digits in
 *             ascending order to form the smallest possible
 *             number (without an invalid leading zero, unless
 *             the number itself is a single digit 0) and
 *             displays it.
 *****/
void generateSmallest(void)
{
    long long num;
    int digits[20], count = 0, i, j, temp;

    printf("\nEnter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return;
    }

    /* Extract digits */
    while (num > 0)
    {
        digits[count++] = num % 10;
        num = num / 10;
    }

    /* Sort digits in ascending order (bubble sort) */
    for (i = 0; i < count - 1; i++)
    {
        for (j = 0; j < count - 1 - i; j++)
        {
            if (digits[j] > digits[j + 1])
            {
                temp = digits[j];
                digits[j] = digits[j + 1];
                digits[j + 1] = temp;
            }
        }
    }

    /* If there is more than one digit and the first digit is 0,
       swap it with the first non-zero digit so the result does
       not have a meaningless leading zero. */
    if (count > 1 && digits[0] == 0)

```

```

    {
        for (i = 1; i < count; i++)
        {
            if (digits[i] != 0)
            {
                temp = digits[0];
                digits[0] = digits[i];
                digits[i] = temp;
                break;
            }
        }
    }

    printf("Smallest number possible : ");
    for (i = 0; i < count; i++)
        printf("%d", digits[i]);
    printf("\n");
}

/*****
 * Function : swapFirstLast
 * Purpose  : Reads a positive integer and swaps its first and
 *             last digits, then displays the resulting number.
 *****/
void swapFirstLast(void)
{
    long long num, original;
    int digits[20], count = 0, i, temp;

    printf("\nEnter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return;
    }

    original = num;

    /* Extract digits (digits[0] = last digit of number) */
    while (num > 0)
    {
        digits[count++] = num % 10;
        num = num / 10;
    }

    /* digits[0] is the last digit, digits[count-1] is the first digit */
    temp = digits[0];
    digits[0] = digits[count - 1];
    digits[count - 1] = temp;

    printf("Original number          : %lld\n", original);
    printf("After swapping first & last digit : ");
    for (i = count - 1; i >= 0; i--)
        printf("%d", digits[i]);
    printf("\n");
}

/*****
 * Function : checkPalindrome

```

```

* Purpose : Reads a positive integer, reverses it and checks
*           whether the reversed number is equal to the
*           original number. Displays whether it is a
*           palindrome or not.
*****/
void checkPalindrome(void)
{
    long long num, original, reverse = 0, remainder;

    printf("\nEnter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return;
    }

    original = num;

    while (num > 0)
    {
        remainder = num % 10;
        reverse = reverse * 10 + remainder;
        num = num / 10;
    }

    printf("Original number : %lld\n", original);
    printf("Reversed number : %lld\n", reverse);

    if (original == reverse)
        printf("%lld is a PALINDROME number.\n", original);
    else
        printf("%lld is NOT a palindrome number.\n", original);
}

```

5. Compilation

The program was compiled using GCC on a Linux terminal with the following command:

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$ gcc -Wall -o digit_ops digit_ops.c
$ ./digit_ops

```

The program compiled successfully with no errors or warnings, confirming correct syntax and use of the switch-case and void function structure.

6. Sample Program Output

Below is a consolidated run of the program exercising all four menu options in sequence (inputs: 3521 for largest, 3521 for smallest, 1234 for digit swap, and 121 for the palindrome check).

```

=====

```



```

DIGIT BASED OPERATIONS ON A POSITIVE INTEGER
=====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digit
4. Check whether the number is a palindrome
5. Exit
-----
Enter your choice (1-5):
Enter a positive integer: Largest number possible : 5321

```

```

=====
DIGIT BASED OPERATIONS ON A POSITIVE INTEGER
=====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digit
4. Check whether the number is a palindrome
5. Exit
-----
Enter your choice (1-5):
Enter a positive integer: Smallest number possible : 1235

```

```

=====
DIGIT BASED OPERATIONS ON A POSITIVE INTEGER
=====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digit
4. Check whether the number is a palindrome
5. Exit
-----
Enter your choice (1-5):
Enter a positive integer: Original number          : 1234
After swapping first & last digit : 4231

```

```

=====
DIGIT BASED OPERATIONS ON A POSITIVE INTEGER
=====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digit
4. Check whether the number is a palindrome
5. Exit
-----
Enter your choice (1-5):
Enter a positive integer: Original number : 121
Reversed number : 121
121 is a PALINDROME number.

```

```

=====
DIGIT BASED OPERATIONS ON A POSITIVE INTEGER
=====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digit
4. Check whether the number is a palindrome
5. Exit
-----
Enter your choice (1-5):
Exiting the program. Thank you!

```

7. Result / Conclusion

The menu-driven C program was successfully written, compiled and executed. All four digit-based operations — generating the largest number, generating the smallest number, swapping the first and last digits, and checking for a palindrome — were implemented using separate void functions and tested against multiple inputs, including edge cases such as single-digit numbers, numbers producing a leading zero, and invalid menu choices. All test cases produced the expected output, confirming that the program works correctly.